Grade: Check Plus

Assignment #1

a.

$P(x \geq 60) = P(60) + P(80) + P(100)$

✓ Qa

$P(x \geq 60) = .15 + .25 + .05$

Probability that a randomly selected customers draws at least $60 = .45

b. Expected value

x	f(x)	x · f(x)
10	.10	1.0
20	.15	3.0
40	.30	12.00
60	.15	9.00
80	.25	20.00
100	.05	5.0
Sum	1.00	

✓ Qb

expected value

$\mu = (10 \times .10) + (20 \times .15) + (40 \times .30) + (60 \times .15) + (80 \times .25) + (100 \times .05) = 50$

expected value is $50

✓ Qd

Variance = square root of the variance

$\sigma = \sqrt{690} \approx 26.2678$ ✓ Qc

c. The manager needs to be aware the expected value is the average over the long term or mean. The more transactions that occur, the closer they will be to the expected.

d. One standard deviation from the mean is $[\mu - 1\sigma, \mu + 1\sigma]$

$P = 50 - 26.27, 50 + 26.27 = [23.73, 76.27]$

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$P(23.73 \leq x \leq 76.27) = .30 + .15$

$P = .45$

✓ Qe

e. Two standard deviations $\Rightarrow [\mu - 2\sigma, \mu + 2\sigma]$

$P = [50 - 52.54, 50 + 52.54]$

$[-2.54, 102.54]$ amounts to withdraw are 10, 20, 40, 60, 80, 100

the sum of all is 1, therfor probability is 1.00